

Ranking Transfers, Thresholds Don't: Unsupervised Test-Time Adaptation for Cross-Corpus Audio Deepfake Detection

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Abstract—Audio deepfake detectors report near-perfect in-domain accuracy but degrade sharply on speech from unseen corpora—different recording conditions, generators, and languages—limiting real-world use. We make a simple empirical observation with large consequences: across corpora, a fine-tuned self-supervised detector’s *ranking* (ROC-AUC) largely transfers even when its decision *threshold* does not calibrate. This means the model’s own most-confident predictions on unlabeled target audio are reliable, which in turn enables stable unsupervised *test-time adaptation* (TTA): we self-train on confident-ranked pseudo-labels and enforce channel-consistency, adapting only LayerNorm and the classifier head—no target labels and no source data at test time. In a leave-one-corpus-out evaluation over three seeds, the method cuts Equal Error Rate (EER) on held-out In-the-Wild from 15.86% to 13.17% (17.0% relative, 3/3 seeds) and on a held-out multilingual target, Arabic audio deepfakes, from 28.93% to 27.18% (3/3 seeds), and it holds up under an inductive check on unseen target clips. In contrast, entropy-minimisation TTA (Tent) collapses to chance on both targets, and train-time domain-adversarial objectives do not help. An ablation shows the two components play different roles depending on the size of the domain gap: confident self-training alone suffices under a moderate shift, while channel-consistency is what rescues adaptation under a larger one.

Index Terms—audio deepfake detection, anti-spoofing, test-time adaptation, cross-corpus generalisation, self-supervised learning, EER.

I. INTRODUCTION

Synthetic speech is realistic enough to threaten voice authentication and enable fraud, motivating reliable spoof/deepfake detectors. On standard benchmarks, self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low error [1]–[3]. Yet these numbers are largely *in-domain*: trained and tested on the same corpus. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6].

We start from a diagnostic measurement. Taking a fine-tuned XLS-R detector and evaluating it on an unseen corpus, we observe that its **ROC-AUC stays high while its accuracy at a fixed threshold collapses**: the model still *rank*s clips correctly (real below fake) but its operating point is miscalibrated for the new domain. This asymmetry is the key that unlocks a practical fix. Because the ranking transfers, the

model’s most-confident predictions on unlabeled target audio are trustworthy, so we can use them to adapt the model to the target domain *at test time*, without any labels.

Our contributions are:

- An empirical finding—*ranking transfers cross-corpus even when the threshold does not*—and its consequence for unsupervised adaptation.
- A simple, stable TTA method: confident-ranked pseudo-label self-training plus channel-consistency, adapting only LayerNorm and the head.
- A leave-one-corpus-out, multilingual evaluation (In-the-Wild, Arabic) with multi-seed, ablation, and inductive analyses, and a clear contrast against entropy-minimisation TTA (which collapses) and train-time domain-adversarial training (which does not help).

II. RELATED WORK

SSL anti-spoofing. Frozen or fine-tuned wav2vec 2.0 / XLS-R / WavLM front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]. These are strong in-domain but corpus-specific. **Domain generalisation (DG) for ADD.** Train-time methods align real and separate fake across source domains (e.g. aggregation-separation [5]) or fuse SSL features [6]. They target the gap at training time and cannot use information about the deployment domain. **Test-time adaptation.** TTA adapts a trained model to an unlabeled target distribution, e.g. entropy minimisation over normalisation parameters [7]. It is widely used in vision but under-explored for audio deepfake detection; we show naive entropy minimisation is unstable here and propose a stable, ranking-driven alternative.

III. METHOD

A. Problem formulation

Let x be a waveform with label $y \in \{0, 1\}$ (0 real, 1 fake). A detector f_θ maps x to frame features, pools them to an embedding $e \in \mathbb{R}^d$, and outputs a fake probability

$$s_\theta(x) = \sigma(w^\top e + b) = P_\theta(y=1 | x), \quad \sigma(z) = \frac{1}{1+e^{-z}}. \quad (1)$$

Training on a labeled *source* corpus $\mathcal{D}_S = \{(x_i, y_i)\}$ minimises the cross-entropy

$$\mathcal{L}_{\text{CE}}(\theta; \mathcal{D}) = -\frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} [y \log s_\theta(x) + (1-y) \log(1-s_\theta(x))]. \quad (2)$$

At test time we receive an *unlabeled* target corpus $\mathcal{D}_T = \{x_j\}_{j=1}^N$ from a shifted distribution (different corpus, channel, or language); no target labels and no source data are available.

B. Detector

The encoder g_ψ is XLS-R-300M with only its top four transformer layers trainable. Given frame features $\{h_t\}_{t=1}^T$, attentive temporal pooling forms the embedding

$$\alpha_t = \frac{\exp(v^\top h_t)}{\sum_{t'} \exp(v^\top h_{t'})}, \quad e = \sum_{t=1}^T \alpha_t h_t, \quad (3)$$

followed by a projection and the linear head of Eq. (1). Fine-tuning only the top layers closes most of the in-domain gap while remaining trainable on one GPU.

C. The ranking–calibration gap

For a threshold τ the decision is $\hat{y}_\tau(x) = \mathbf{1}[s_\theta(x) \geq \tau]$, whose accuracy depends on τ . Ranking quality, by contrast, is threshold-free:

$$\text{AUC} = \Pr_{x^+ \sim \text{fake}, x^- \sim \text{real}} [s_\theta(x^+) > s_\theta(x^-)], \quad (4)$$

which is invariant to any strictly increasing reparametrisation of s_θ . Empirically, under the source→target shift AUC degrades far less than accuracy at a fixed threshold does (Fig. 1: AUC 0.899 yet accuracy at $\tau=0.5$ is only 76% on this fold). *Ranking transfers more than calibration does.* We therefore trust the score *order*, not its absolute value: the extreme quantiles are higher-precision pseudo-labels than a fixed threshold would give, even when s_θ is mis-calibrated.

D. Test-time adaptation

Let Q_α denote the α -quantile of the target scores $\{s_\theta(x_j)\}_{j=1}^N$. Each adaptation epoch we (re-)assign confident pseudo-labels from the extreme tails,

$$\tilde{y}_j = \begin{cases} 0, & s_\theta(x_j) \leq Q_q, \\ 1, & s_\theta(x_j) \geq Q_{1-q}, \\ \emptyset, & \text{otherwise (ignored),} \end{cases} \quad (5)$$

and let $\mathcal{C} = \{j : \tilde{y}_j \neq \emptyset\}$. Self-training is cross-entropy on this confident set,

$$\mathcal{L}_{\text{ST}} = -\frac{1}{|\mathcal{C}|} \sum_{j \in \mathcal{C}} [\tilde{y}_j \log s_\theta(x_j) + (1-\tilde{y}_j) \log(1-s_\theta(x_j))]. \quad (6)$$

With a stochastic channel augmentation $a(\cdot)$ (random gain + additive noise) and $p_\theta(x) = [1-s_\theta(x), s_\theta(x)]$, a consistency term enforces prediction invariance to the channel,

$$\mathcal{L}_{\text{cons}} = \frac{1}{N} \sum_{j=1}^N \|p_\theta(a(x_j)) - \text{sg}[p_\theta(x_j)]\|_2^2, \quad (7)$$

Algorithm 1 Real-manifold test-time adaptation (per target corpus)

Require: source model f_θ ; unlabeled target $\mathcal{D}_T = \{x_j\}_{j=1}^N$; quantile q ; weight λ ; epochs E ; augmentation $a(\cdot)$; step size η

Ensure: adapted model f_θ

- 1: make top-layer LayerNorm $\{\gamma, \beta\}$ and head ϕ trainable; freeze all else
 - 2: **for** $e = 1, \dots, E$ **do**
 - 3: $s_j \leftarrow s_\theta(x_j) \quad \forall j$ ▷ score the whole target set
 - 4: $Q_q, Q_{1-q} \leftarrow$ quantiles of $\{s_j\}_{j=1}^N$
 - 5: $\tilde{y}_j \leftarrow 0$ if $s_j \leq Q_q$; 1 if $s_j \geq Q_{1-q}$; else $\emptyset$
 - 6: $\mathcal{C} \leftarrow \{j : \tilde{y}_j \neq \emptyset\}$ ▷ confident tails
 - 7: **for** minibatch $\mathcal{B} \subset \mathcal{D}_T$ **do**
 - 8: $\mathcal{L}_{\text{ST}} \leftarrow$ CE over $\mathcal{B} \cap \mathcal{C}$ with targets $\tilde{y}$ ▷ Eq. (6)
 - 9: $\mathcal{L}_{\text{cons}} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} \|p_\theta(a(x)) - \text{sg}[p_\theta(x)]\|_2^2$
 - 10: $\theta \leftarrow \theta - \eta \nabla_\theta (\mathcal{L}_{\text{ST}} + \lambda \mathcal{L}_{\text{cons}})$
 - 11: **end for**
 - 12: **end for**
-

where $\text{sg}[\cdot]$ is stop-gradient. The adaptation objective is

$$\min_{\{\gamma, \beta\}, \phi} \mathcal{L}_{\text{ST}} + \lambda \mathcal{L}_{\text{cons}}, \quad \lambda = 0.3, \quad q = 0.3, \quad (8)$$

optimising *only* the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned layers and the head ϕ ; the rest of ψ stays frozen. Restricting the trainable set and using only confident tails prevents the degenerate collapse $s_\theta(\cdot) \rightarrow \text{const}$ to which entropy minimisation is prone, as we confirm empirically against Tent (Sec. V). Algorithm 1 summarises the procedure.

IV. EXPERIMENTAL SETUP

Protocol. Leave-one-corpus-out. The source pool merges ASVspoof2019-LA, a multi-generator TTS set (eight distinct synthesis systems), and every corpus *except* the held-out target; source fakes are water-filled across corpora and generators so no single generator dominates training. **Targets (unseen, held out entirely from source training).** In-the-Wild (English, real+fake) and the Arabic Audio Deepfake set (ArAD), our multilingual target. **Metric.** Equal Error Rate (EER) and ROC-AUC. With false-positive and false-negative rates $\text{FPR}(\tau), \text{FNR}(\tau)$,

$$\text{EER} = \text{FPR}(\tau^*) = \text{FNR}(\tau^*), \quad \tau^* : \text{FPR}(\tau^*) = \text{FNR}(\tau^*), \quad (9)$$

which is threshold-free and thus insensitive to the calibration drift above. **Setting.** Transductive TTA (adapt on the unlabeled target pool, then score it); we additionally report an inductive check (adapt on one half of the target, test on the disjoint half). **Implementation.** 16 kHz audio cached at 4 s and randomly cropped to 3 s at train time (label-preserving gain/noise augmentation); bf16; Adam, lr 10^{-4} ; top-4-layer XLS-R fine-tuning; source trained 6 epochs, batch 64; single GPU. All results are mean $\pm$ std over three seeds (0, 1, 2), each reported as “improved in $k/3$ seeds” rather than a significance claim, since $n=3$ does not support one.

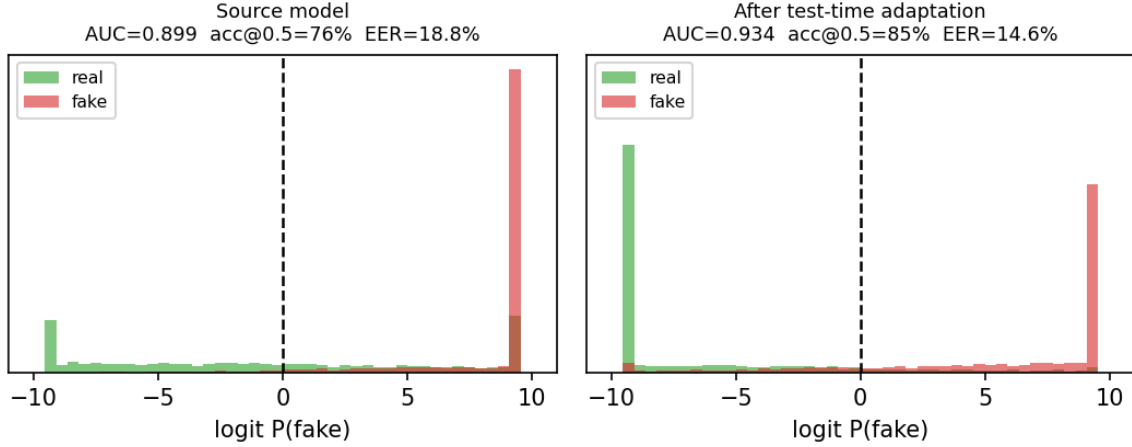


Fig. 1. Detector scores (logit axis; scores concentrate near the extremes) for real vs. fake clips on the *unseen* In-the-Wild corpus (representative seed). **Left:** the source model pushes many real clips past the 0.5 threshold ($\text{acc@0.5}=76\%$) yet still ranks most real below fake ($\text{AUC}=0.899$). **Right:** unsupervised test-time adaptation shifts the real mass back across the threshold ($\text{acc@0.5}=85\%$, $\text{AUC}=0.934$, $\text{EER } 18.8\% \rightarrow 14.6\%$ on this fold). Table I reports the 3-seed mean.

TABLE I
CROSS-CORPUS EER % / AUC, MEAN $\pm$ STD OVER 3 SEEDS
(LEAVE-ONE-CORPUS-OUT; LOWER EER BETTER).

Method	In-the-Wild	Arabic (ArAD)
Source-only	15.86 $\pm$ 2.84 / .922	28.93 $\pm$ 0.92 / .784
Tent [7]	43.65 $\pm$ 9.41 / .564	43.82 $\pm$ 7.18 / .579
Ours (TTA)	13.17$\pm$1.63 / .945	27.18$\pm$0.97 / .805

TABLE II
SELF-TRAINING ALONE VS. WITH CONSISTENCY, MEAN OVER 3 SEEDS
(EER % / AUC).

Configuration	In-the-Wild	Arabic
Source-only	15.86 / .922	28.93 / .784
Self-training only	14.62 / .931	28.89 / .787
+ consistency (ours)	13.17 / .945	27.18 / .805

V. RESULTS

A. Cross-corpus adaptation

Table I is the main result, leave-one-corpus-out, mean $\pm$ std over three seeds. On In-the-Wild, TTA reduces EER from $15.86 \pm 2.84\%$ to $13.17 \pm 1.63\%$ (a 17.0% relative reduction) and raises AUC $0.922 \rightarrow 0.945$; the improvement holds in 3/3 seeds (paired mean -2.69 EER). On Arabic, the multilingual target, TTA reduces EER from $28.93 \pm 0.92\%$ to $27.18 \pm 0.97\%$ (6.0% relative; 3/3 seeds, paired mean -1.75). Vanilla entropy-minimisation TTA (Tent) *collapses* on both targets—EER above 43% and AUC near chance (0.56–0.58), consistently across all three seeds—underlining that the confident-pseudo-label design, not TTA in general, is what makes adaptation stable here.

B. Ablation: what drives the gain

Table II adds self-training alone (no consistency) at each target, using the same leave-one-corpus-out checkpoints. The two targets tell different stories. On In-the-Wild (smaller domain gap, source AUC 0.922), self-training alone recovers most of the improvement ($15.86 \rightarrow 14.62$), with consistency adding a further increment. On Arabic (larger gap, source AUC 0.784), self-training alone is *not distinguishable from source-only* ($28.93 \rightarrow 28.89$, no seed improved), and the entire gain comes from adding channel-consistency ($\rightarrow 27.18$). This

suggests confident-tail pseudo-labels alone become unreliable as the domain shift grows, and consistency regularisation is what keeps adaptation stable under a larger shift—not merely an add-on. A real-anchoring contrastive term (pulling target-real embeddings to a source real prototype), tested in earlier pilots, did not help in either regime and is not included here.

C. Hyperparameter sensitivity

We sweep the three adaptation knobs—confident-tail quantile q , adaptation epochs E , and consistency weight λ —one at a time around the defaults ($q=0.3$, $E=4$, $\lambda=0.3$) used everywhere else in the paper, on seed 0 of both targets (Table III).¹ Two consistent findings. First, *more adaptation epochs help on both targets* ($E=8$ beats $E=4$ by 1.2 points on In-the-Wild and 1.6 on Arabic): our default of 4 epochs was a conservative, compute-matched choice for the main grid, not a tuned optimum, and a deployment that can afford more adaptation steps should expect a further gain. Second, *consistency weight $\lambda>0$ is uniformly necessary*: setting $\lambda=0$ (self-training only, no consistency) is the worst configuration in the sweep on both targets, matching the ablation in Table II.

¹An earlier sweep shared one continuous RNG stream across configurations, so adjacent points differed partly by batch-order noise rather than the knob alone; every point reported here re-seeds `torch.manual_seed(0)` immediately before adaptation, isolating the knob’s effect.

TABLE III
HYPERPARAMETER SWEEP (SEED 0, EER % / AUC; DEFAULT ROW
 $q=0.3$, $E=4$, $\lambda=0.3$ IN GREEN).

Setting	In-the-Wild	Arabic
Source-only	18.73 / .900	28.56 / .781
$q = 0.1$	14.87 / .933	25.14 / .825
$q = 0.2$	14.70 / .933	23.05 / .849
$q = 0.3$ (default)	14.25 / .934	24.74 / .836
$q = 0.4$	14.47 / .939	24.51 / .830
$E = 2$	15.38 / .923	26.99 / .809
$E = 4$ (default)	14.25 / .934	24.74 / .836
$E = 8$	13.03 / .945	23.12 / .852
$\lambda = 0$ (no consistency)	15.75 / .919	25.46 / .822
$\lambda = 0.3$ (default)	14.25 / .934	24.74 / .836
$\lambda = 1.0$	13.98 / .937	25.28 / .832

The optimal confident-tail quantile, in contrast, is *target-dependent* ($q=0.3$ on In-the-Wild vs. $q=0.2$ on Arabic, the larger-shift target), suggesting the fraction of trustworthy tail predictions shrinks somewhat as the domain gap grows—consistent with the mechanism story of Sec. V-B. We keep one fixed configuration for the whole paper for a fair cross-target comparison rather than tuning per target.

D. Inductive check

To rule out transductive memorisation, we adapt on one half of each target and evaluate on the disjoint half. On In-the-Wild, EER improves $15.72 \pm 2.76\% \rightarrow 14.27 \pm 2.94\%$, holding in 3/3 seeds (paired mean -1.46)—the adaptation captures domain-level structure that transfers to unseen target clips, not just the specific evaluation items. On Arabic the inductive gain is smaller and less consistent ($28.53 \pm 0.54\% \rightarrow 27.85 \pm 1.68\%$, 2/3 seeds, paired mean -0.68); we report this honestly rather than only the more favourable transductive number.

E. Embedding geometry: class organises the space, not domain

To probe *why* ranking transfers across corpora (Sec. III-C), we visualise 500 source and 500 In-the-Wild target embeddings with t-SNE (Fig. 2) and train a linear probe to predict source-vs-target origin from the frozen embedding. Two observations. First, in both the pre-adaptation and post-adaptation embedding space, source and target points are thoroughly *intermixed within* each real/fake cluster rather than forming separate domain islands—the representation is already organised primarily by *class*, which is consistent with the ranking-transfer finding: if the embedding mainly encoded corpus identity, cross-corpus ranking could not survive the shift. Second, the domain probe is only modestly weaker after adaptation ($80.3 \pm 1.9\% \rightarrow 78.3 \pm 2.7\%$, within one standard deviation of no change)—adaptation does *not* erase corpus identity from the representation. This distinguishes our mechanism from adversarial domain-invariance approaches (e.g. gradient-reversal on domain identity, which we tried and which destabilised training, Sec. V-E): the improvement here



Fig. 2. t-SNE of 500 source + 500 In-the-Wild target embeddings, before/after adaptation (real=green, fake=red; circle=source, triangle=target). Source and target intermix within each class cluster in both panels; the linear domain-origin probe changes only marginally (80%→78%), indicating the gain comes from decision-boundary reshaping, not domain erasure.

comes from re-shaping the decision boundary using rank-reliable pseudo-labels within an already class-organised space, not from removing domain information.

F. What did not work

Train-time domain-adversarial training (gradient reversal on corpus identity) plus a real-anchored contrastive loss—our initial approach—did not improve cross-corpus EER and was unstable (adversarial training collapsed). This negative result motivated the shift to test-time adaptation and is reported for transparency; the embedding-geometry analysis above (Sec. V-D) offers a post-hoc explanation: the representation was never organised primarily by domain in the first place, so explicitly adversarially erasing domain identity had little useful signal to act on and mainly destabilised optimisation.

VI. DISCUSSION AND LIMITATIONS

The method is unsupervised in labels but *transductive*: it uses unlabeled target inputs at adaptation time, a standard TTA setting that we state explicitly; the inductive check addresses the memorisation concern, though it is weaker on Arabic than on In-the-Wild (2/3 vs. 3/3 seeds), which we report rather than suppress. Source-model quality itself is not perfectly stable under our training recipe: In-the-Wild source EER ranges 13.1–18.8% across seeds (± 2.84), a wider spread than we would like, and future work should reduce this variance (e.g. smaller batches or a tuned learning-rate schedule) before scaling to a larger seed count. Encouragingly, the relative ordering $\text{EER}(\text{ours}) < \text{EER}(\text{source}) \ll \text{EER}(\text{Tent})$ holds in every seed regardless of this variance. The approach is cheap (adapt a trained model in minutes) and orthogonal to the choice of SSL back-end.

VII. CONCLUSION

Cross-corpus audio deepfake detection fails not because ranking is lost but because the operating point is miscalibrated. Exploiting this, a simple unsupervised test-time adaptation—confident-ranked self-training plus channel-consistency—reliably improves generalisation to unseen corpora and languages, where entropy-minimisation TTA col-

lapses and train-time domain-adversarial training does not help.

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